MIL-B-11845B(ME)
5 April 1974
SUPERSEDING
MIL-B-11845A(CE)
22 May 1958

MILITARY SPECIFICATION

BROMOTHYMOL BLUE SOLUTION:

INDICATOR, 6.0 TO 7.6 pH RANGE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers bromothymol blue, indicator solution for indicating a pH change from 6.0 or less to 7.6 or vice versa of initially colorless aqueous solutions.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-300 - Chemicals, Liquid: Packaging and Packing of.

FSC 6820

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STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. Bromothymol indicator solution shall contain 0.4 gram of bromothymol blue (3',3"-Dibromothymolsulfonephthalein) per liter of a distilled water and alcohol solution (50 percent water, 50 percent methanol by volume) adjusted to not less than 6.5 nor more than 7.1 pH by use of reagent grade hydrochloric acid, sodium hydroxide, or both. The solution shall be free from suspended matter or other material which may affect the performance of the indicator.

3.2 Colors. When tested as specified in 4.4.2.1, the solutions shall be colored as follows:

pH	Color
6.0	Yellow
6.8	Green
7.6	Blue

3.3 Identification marking. The label for the bottle shall contain the words: Bromothymol Blue Solution: Indicator; the color, the pH number, and the letters US.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, Inspection Level S-4.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the defects specified in 4.4.1. AQL shall be 2.5 defective.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2 AQL shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. Bromothymol blue solution shall be examined as specified herein for the following defects:

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Major

101. Suspended matter or foreign material in the solution.
102. Container cracked or broken.
103. Identification marking missing or not as specified.

4.4.2 Tests.

4.4.2.1 Color. Four drops of bromothymol blue shall be added to 10 ml of each of the following buffer solutions:

pH	Buffer solutions
6.0	5.6 ml of 0.1 N sodium hydroxide and 50.00 ml of 0.1 M monopotassium phosphate diluted to 100 ml.
6.8	22.4 ml of 0.1 N sodium hydroxide and 50.00 ml of 0.1 M monopotassium phosphate diluted to 100 ml.
7.6	38.5 ml of 0.1 N sodium hydroxide and 50.00 ml of 0.1 M monopotassium phosphate diluted to 100 ml.

Color of solutions not as specified in 3.2 shall constitute failure of this test.

4.5 Inspection of preparation for delivery. The packaging, packing, and marking of the solution shall be examined to determine compliance with PPP-C-300.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. The packaging, packing, and marking of the bromothymol blue solution shall be in accordance with PPP-C-300, level as specified (see 6.2). Unless otherwise specified (see 6.2), the solution shall be packaged in unit quantities of 500 ml plus 10 minus 0 ml with the containers conforming to PPP-C-300, Type III, Class 1, and intermediate packaged as specified for one pint containers in PPP-C-300.

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6. NOTES

6.1 Intended use. The indicator solution is intended to be used to determine the pH of water during the water purification process.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and level of packing required (see 5.1).
(c) Packaging size if other than as specified (see 5.1).

Custodian:

Army - ME

Preparing activity:

Army - ME

Project No. 6820-A056

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-B-11845B(ME) Bromothymol Blue Solution: Indicator, 6.0 to 7.6 pH Range

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426
11 Oct 72 (Rev)

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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